Discussion Unit: DSE Analytics, Lambda Architecture

- Discuss Lamdba architecture as it relates to DSE, DSE Analytics

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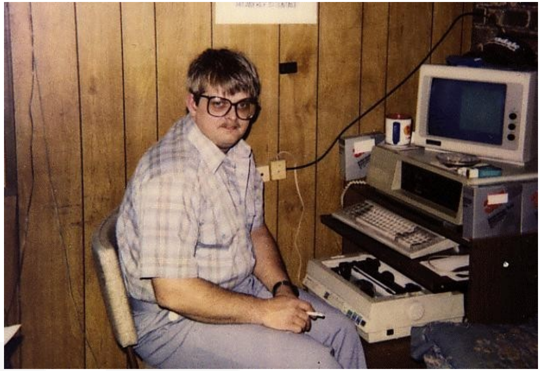
Discussion Lab:

Matching pairs – Match the attributes on the right with the areas on the left

Architectures of years past and present-

DSE Analytics: Architectures of years past



- Data Mart
- Data Warehouse
- Enterprise Data Warehouse/Layer
- Data Lake
- Excel Spreadsheet

End of Discussion Lab:

Wikipedia: Lambda Architecture

Lambda architecture is a data-processing architecture designed to handle massive quantities of data by taking advantage of both batch and stream-processing methods. This approach to architecture attempts to balance latency, throughput, and fault-tolerance by using batch processing to provide comprehensive and accurate views of batch data, while simultaneously using real-time stream processing to provide views of online data. The two view outputs may be joined before presentation. The rise of lambda architecture is correlated with the growth of big data, real-time analytics, and the drive to mitigate the latencies of map-reduce.

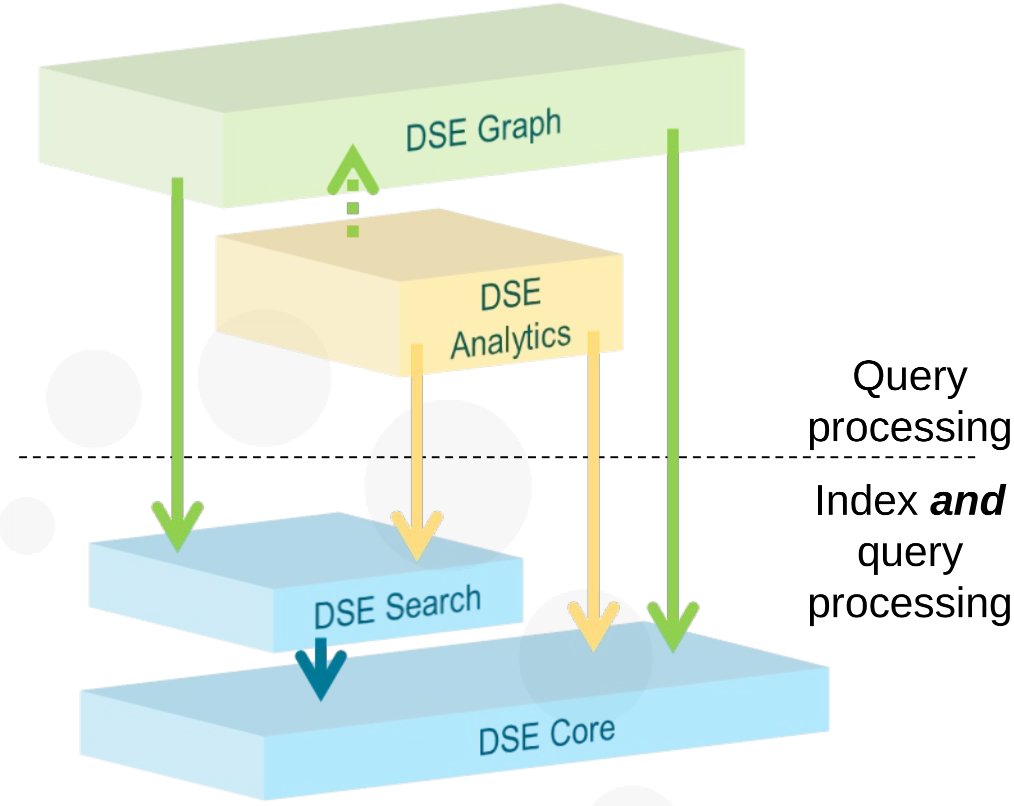
Lambda architecture depends on a data model with an append-only, immutable data source that serves as a system of record.

It is intended for ingesting and processing times tamped events that are appended to existing events rather than overwriting them. State is determined from the natural time-based ordering of the data.

WIKIPEDIA
The Free Encyclopedia

Source: https://en.wikipedia.org/wiki/Lambda_architecture

Review: The 4 Primary Functional Areas to DSE



- All 4 primary functional areas provide query processing

The Why

- DSE Analytics
  - -- *Parallel* Query Processing
    - Horizontal scaling
    - High Speed
  - -- Batch, Streaming, Iterative, Interactive

Review: DSE Analytics, 5 Major Functional Areas

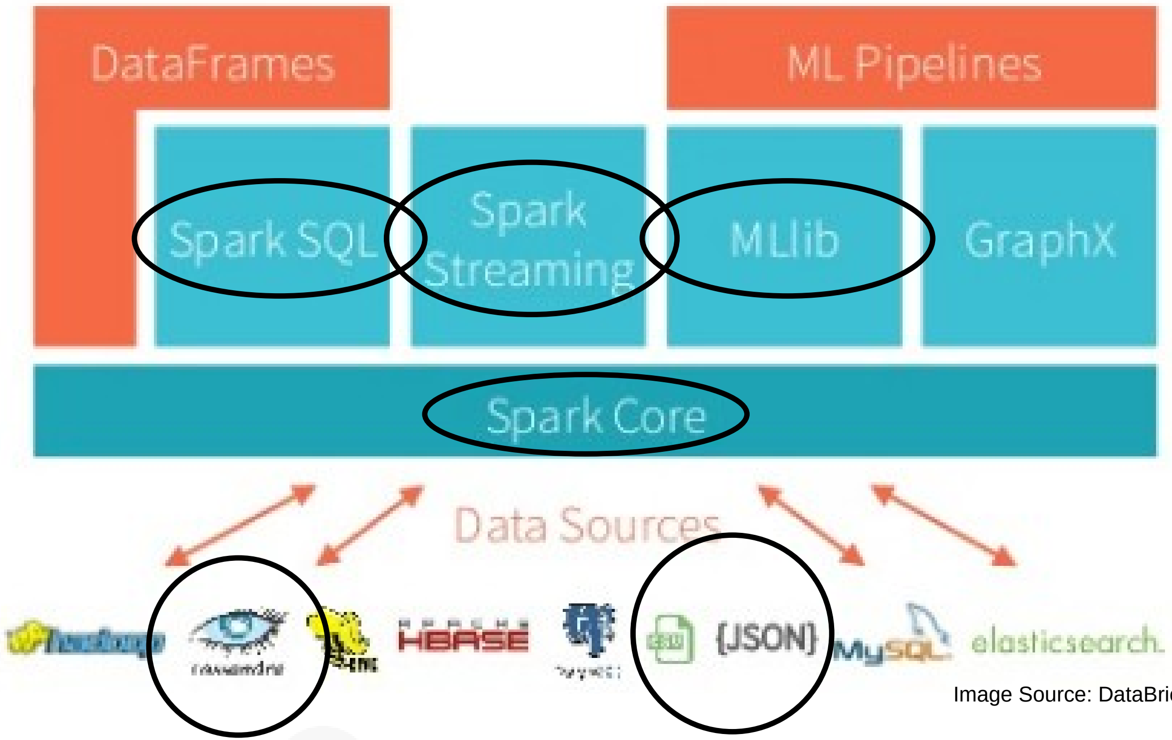


Image Source: DataBricks.com

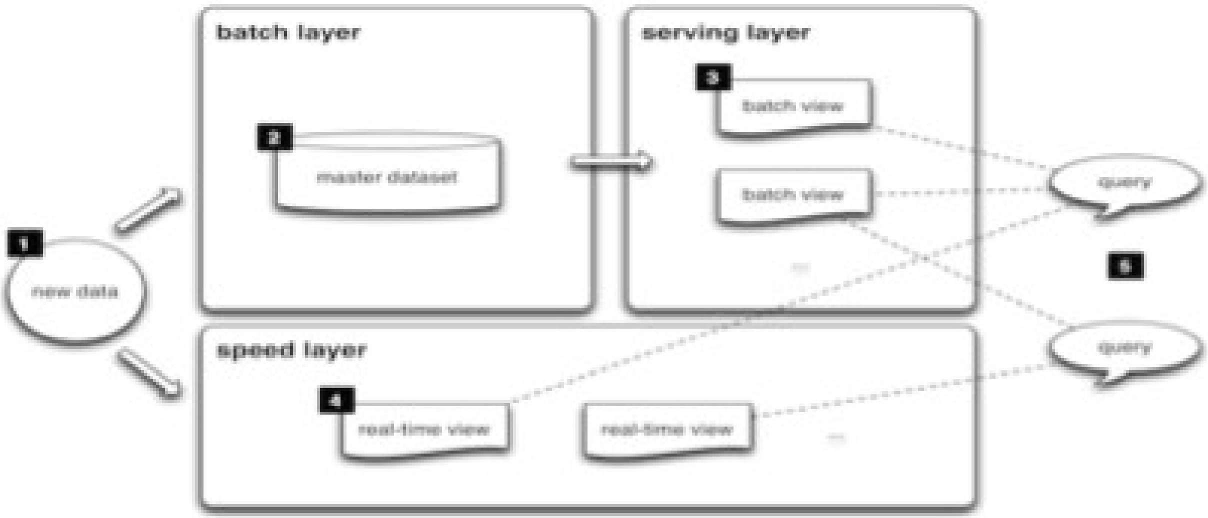
DSE Analytics: Lambda Architecture

DSE Analytics is part of a Lambda Architecture

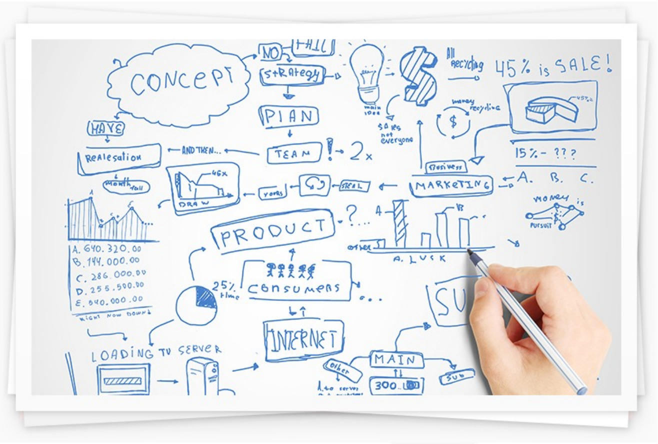
- The idea and original term *Lambda Architecture* was first introduced in Book https://www.manning.com/books/big-data , by Nathan Marz (2012)
- Lambda has become an overloaded phrase; buzzword and marking material without necessarily tying back to the original intent
- True Lambda architectural design in your enterprise applications is considered a key part of an enterprise ready system, making the idea of saying your design is part of Lambda appealing
- Lambda architecture is a data-processing architecture designed to handle massive quantities of data by taking advantage of both batch- and stream-processing methods

DSE Analytics: Lambda Architecture

The lambda architecture is an attempt to solve the problem of computing arbitrary functions on arbitrary data in real time



DSE Analytics: Lambda Architecture



The promise of Lambda:

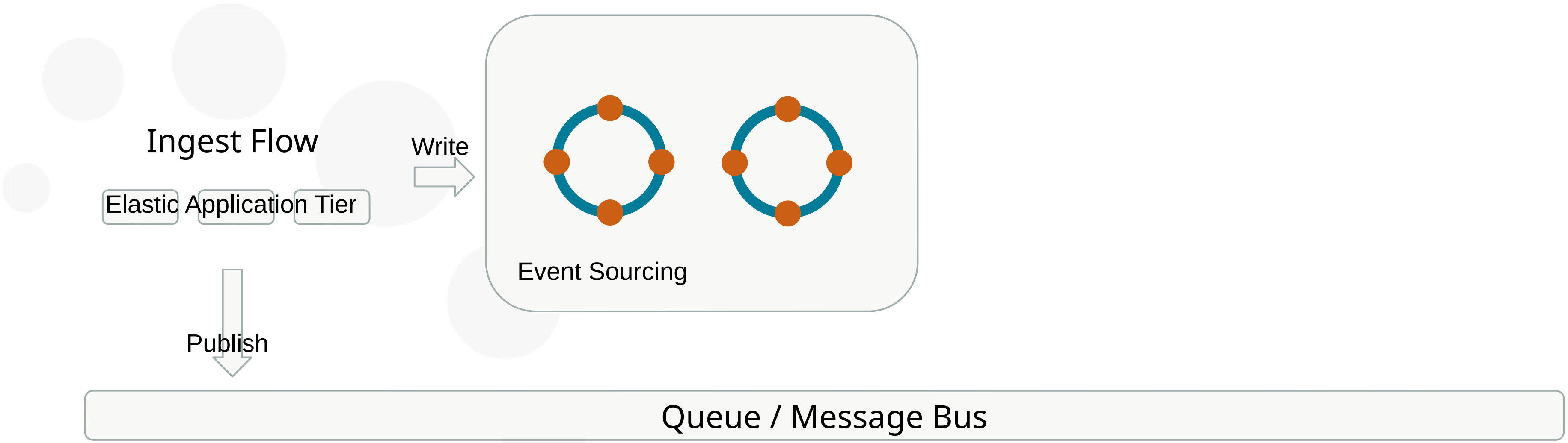
- Analysis happens on streaming ingest allowing for time sensitive value to be captured
- Batch layer re-calculates fast analytics and corrects accuracy
- Serving layer accesses both layers and treats batch layer as authoritative for conflicts

Three layers:

- Streaming/Speed
- Batch
- Serving

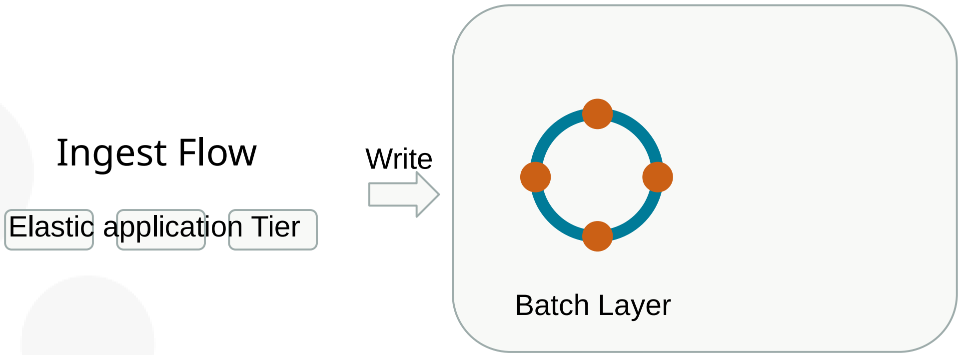
DSE Analytics: Lambda Architecture

Ingest flow - double writes into queue and DSE



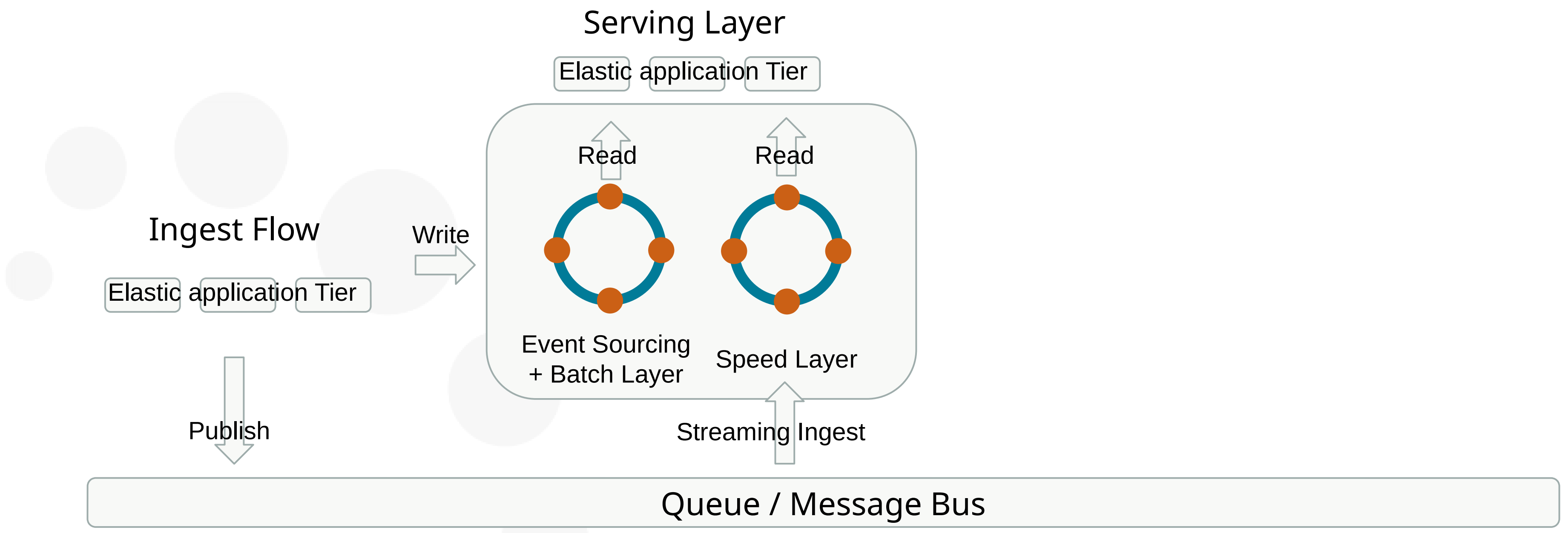
DSE Analytics: Lambda Architecture

Batch Layer - DSE batch aggregations



DSE Analytics: Lambda Architecture

Server Layer - Batch wins for accuracy

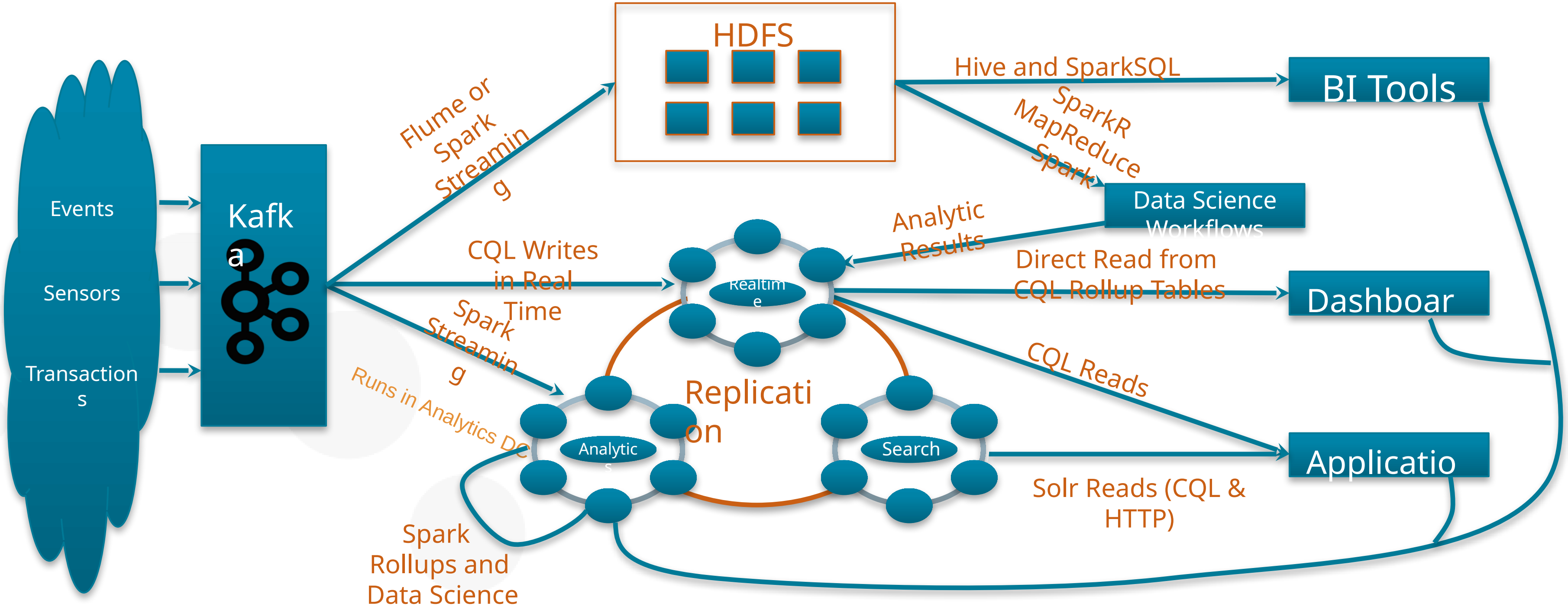


DSE Analytics: Lambda Architecture

Decompose the problem into three layers:

- Batch layer
  -- Stores immutable Master Data Set
  -- Computes Arbitrary Views - functions on data
- Serving layer
  -- Random Access to Batch Views
  -- Updated by batch layer
- Streaming/speed layer
  -- Compensates for high latency of update to serving layer
  -- Fast incremental Algorithms
- Batch layer eventually overrides speed layer
  -- Batch layer looks at all data
  -- Streaming/speed layer for incremental updates

DSE Analytics: Lambda Architecture, How DSE fits



End of Unit: